

A **SIGNAL** IS AN ACT, GESTURE, EVENT OR SOMETHING THAT MAY HAVE BEEN AGREED ON AND WHICH CARRIES INFORMATION.

A SIGNAL IS A FUNCTION WHICH DESCRIBES THE VARIATIONS OF A DETECTABLE QUANTITY BY WHICH INFORMATION CAN BE TRANSMITTED FROM A SOURCE TO A RECEIVER.

SIGNAL CLASSIFICATION

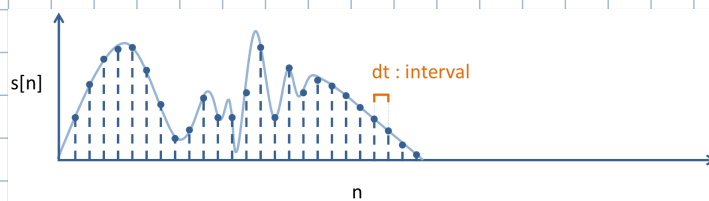
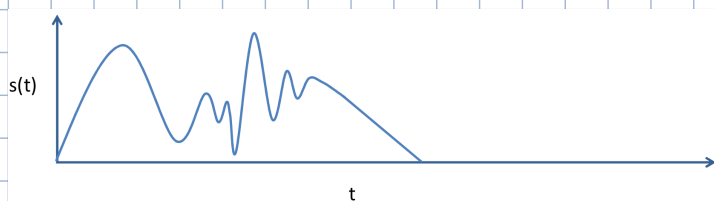
SIGNALS CAN BE CLASSIFIED IN MULTIPLE WAYS:

- NATURAL OR ARTIFICIAL
- CONTINUOUS OR DISCRETE
- ANALOG OR DIGITAL
- PERIODIC OR NON-PERIODIC
- ENERGY OR POWER SIGNALS
- RANDOM OR DETERMINISTIC

CONTINUOUS VS. DISCRETE

A SIGNAL IS **CONTINUOUS** IF IT EXISTS FOR EVERY VALUE OF THE GIVEN VARIABLE(S). IT IS INSTEAD **DISCRETE** IF IT EXISTS ONLY FOR A SPECIFIC SET OF VALUES.

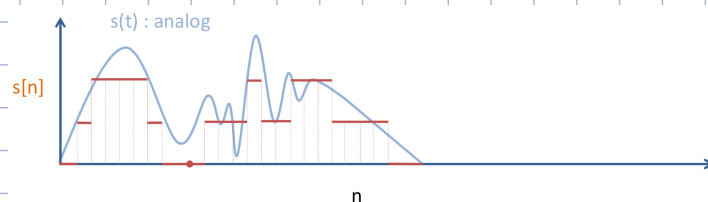
DISCRETE SIGNALS ARE REPRESENTED AS $s[n]$, $n \in \mathbb{Z}$, USING SQUARE BRACKETS.



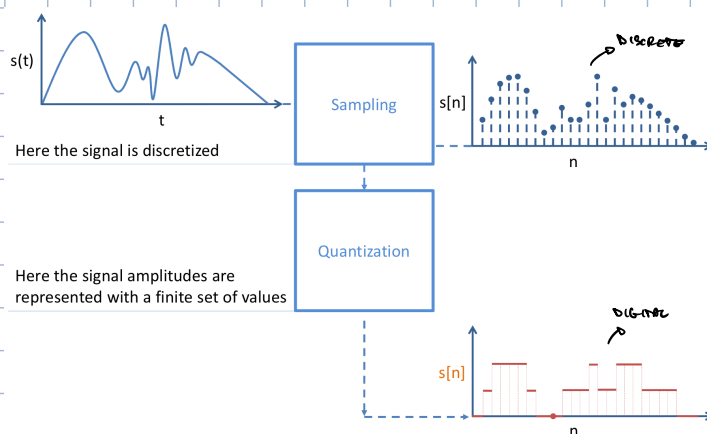
ANALOG VS. DIGITAL

A SIGNAL IS **DIGITAL** IF IT IS **DISCRETE** AND ONLY A **FINITE SET OF VALUES** CAN BE REPRESENTED.

IT IS INSTEAD **ANALOG** IF IT IS **CONTINUOUS** AND ITS AMPLITUDE CAN ASSUME ANY VALUE.



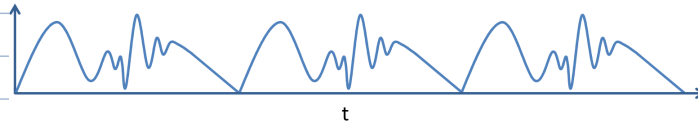
ADC



PERIODIC VS. NON-PERIODIC

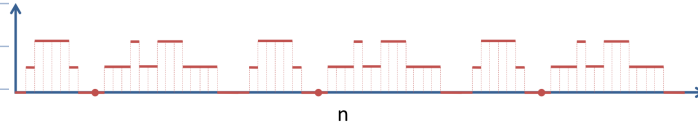
$$s(t) = s(t+nT)$$

with n being an integer and T the period



$$s[n] = s[n+kM]$$

with k being an integer and M being an integer and the period



ENERGY AND POWER SIGNALS

$$E = \sum_{-\infty}^{+\infty} |s[n]|^2$$

$$P_{\infty} = \lim_{N \rightarrow \infty} \frac{1}{2N+1} \sum_{-N}^N |s[n]|^2$$

IF THE ENERGY IS FINITE ($E < \infty$), THE SIGNAL IS AN ENERGY SIGNAL.

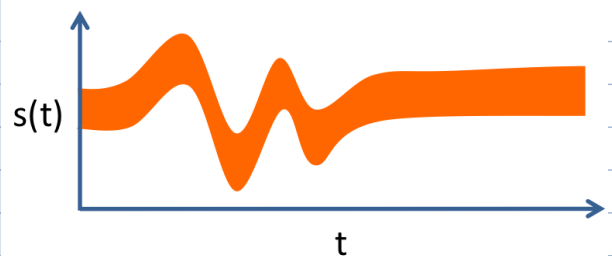
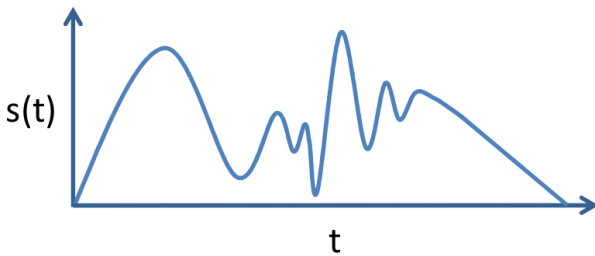
IF THE INFINITE AVERAGE POWER IS FINITE AND NOT ZERO ($P_{\infty} < \infty, \neq 0$), THE SIGNAL IS A POWER SIGNAL.

OTHERWISE IT'S OF NEITHER TYPE.

DETERMINISTIC VS. RANDOM

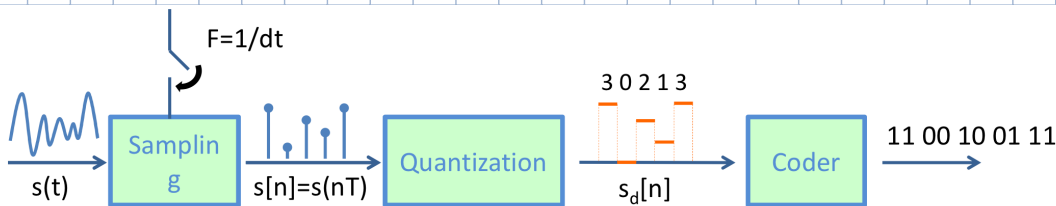
A SIGNAL IS DETERMINISTIC IF IT CAN ASSUME ONE AND ONLY ONE VALUE FOR A GIVEN VALUE OF THE REFERENCE VALUE. IF INSTEAD IT CAN ASSUME A RANGE OF VALUES WITH A CERTAIN PROBABILITY, IT IS RANDOM.

TEST TEST TEST TEST



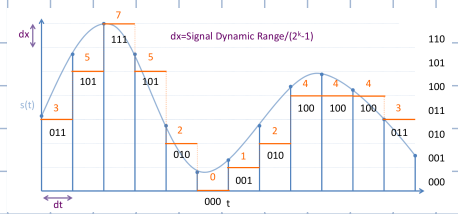
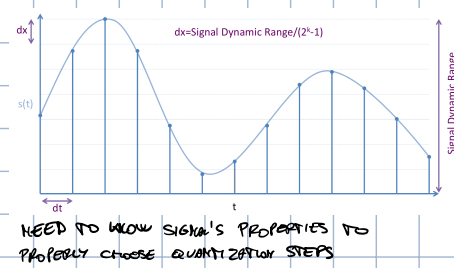
ADC

NEED SOME WAY TO GO FROM AN ANALOG SIGNAL TO A DIGITAL ONE.



TIME - CONTINUOUS $s(t)$
TO
TIME - DISCRETE $s[n]$

DOMAIN - CONTINUOUS
TO
DOMAIN - DISCRETE



QUANTIZATION IS LOSSY $\rightarrow$ THE LARGER THE QUANTIZATION STEP, THE MORE INFORMATION IS LOST.
 $\hookrightarrow$ TRADEOFF FIDELITY / MEMORY

QUANTIZATION ERROR

DYNAMIC RANGE: $DR = \max\{s(t)\} - \min\{s(t)\}$

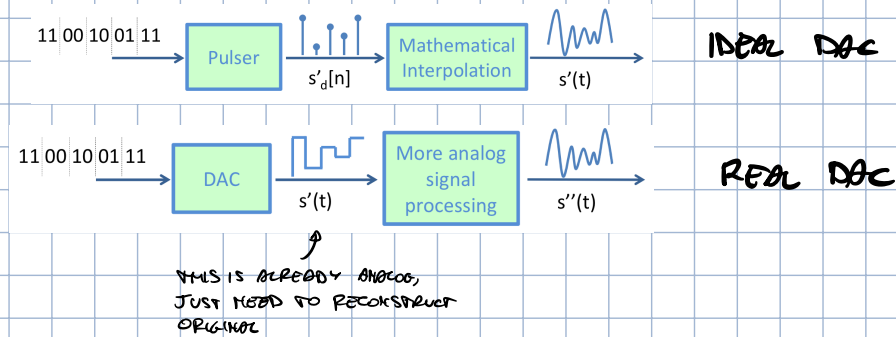
QUANTIZATION STEP: $\frac{DR}{2^n - 1}$

$\rightarrow 0 < \text{ERROR} < \frac{DR}{2^n - 1}$

- Higher DR $\Rightarrow$ Higher Error
- Higher # Bits $\Rightarrow$ Lower Error

DAC

ALSO NEED A WAY TO GO BACK TO ANALOG DOMAIN



DIGITAL $\gg$ ANALOG: TRANSFORMATION

- WEAR
- FLEXIBILITY
- STABILITY